

Document 4 -- Additional Information

Aran Technologies | aranrobotics@gmail.com | IAF MBC-3 Competition

1. Phase I Testing -- Completed

All Phase I systems have been tested and verified. The five-drone swarm simulation on ROS2 Jazzy and Gazebo Harmonic has demonstrated all core MBC-3 requirements: real-time ASP generation, bully-protocol leader election, graceful degradation on drone loss, and sector reassignment -- 344 radar tracks logged with four surviving drones sustaining full mission continuity.

A pre-recorded 4-minute single-drone ISR mission video (1920x1080, v12-MPC-v5) is ready for Phase I presentation: 11-waypoint survey, primary target orbit locked at 50 m radius +/-0.5 m, RTL, and 3D occupancy map save -- verified clean exit. Broadband and drone-to-ground communication links have been tested and confirmed operational.

2. Indigenisation Breakdown

Sub-system	Indigenous Content
Hexacopter airframe	Indigenous design and local fabrication
Antenna panels (6 x per drone)	Indigenous PCB design
AI software stack (CFAR, RF, LLM)	100% indigenous (Python / C++, open-source models)
GCS dashboard (Flask / SocketIO)	100% indigenous
ROS2 swarm coordination stack	100% indigenous
Flight controller firmware	Custom STM32 PCB -- designed from scratch
Compute module (Jetson)	COTS -- imported
Radar frontend (AWR1843)	COTS -- imported
Mesh radio (Doodle Labs)	COTS -- imported

Estimated indigenisation: >= 55% by mission-criticality weighting -- meets MBC-3 sec.2.25 requirement.

3. Team

Aran Technologies is a three-member engineering startup building indigenous defence UAS systems from the ground up. Kishore Udhayakumar | Founder & CEO -- Mechanical Engineering. Leads product development, mechanical design, thermal engineering, and business strategy. Drives go-to-market planning and direct engagement with Indian defence end-users.

Vegi Tejo Bhargav | Co-Founder & CTO -- Mechanical Engineering. Manages project execution, hexacopter airframe fabrication, additive manufacturing, procurement, and hardware integration roadmap.

L. Harsha Vardhan Naidu | Product Head, Electronics -- EEE. Designed the complete autonomous mission stack: PX4 SITL pipeline, ROS2 swarm middleware, GCS dashboard, STM32 flight controller firmware, and three-layer onboard AI pipeline (CFAR -> Random Forest -> LLM).

Point of Contact: aranrobotics@gmail.com | +91 72888 40612

4. Phase I Readiness

Simulation stack, GCS dashboard, graceful degradation demonstration, and pre-recorded mission video are complete and ready for live presentation at New Delhi, 13-24 July 2026.